

Лабораторная Работа №9. Использование протокола STP. Агрегирование каналов

Сетевой администрирование

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Изучение возможностей протокола STP и его модификаций по обеспечению отказоустойчивости сети, агрегированию интерфейсов и перераспределению нагрузки между ними.

Логическая схема локальной сети с резервным соединением

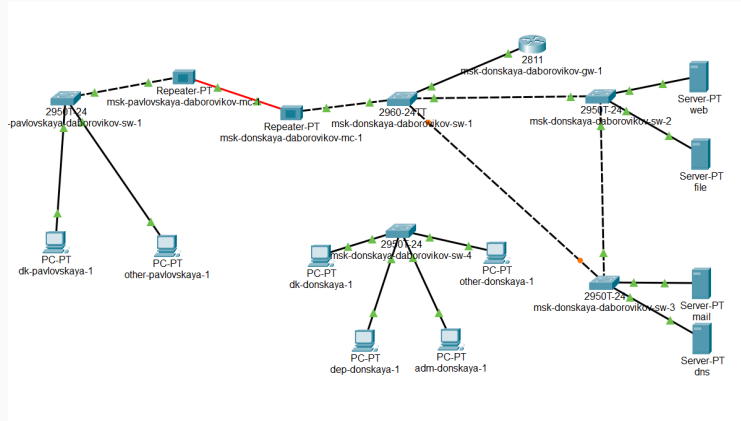


Figure 1: Логическая схема локальной сети с резервным соединением

```
msk-donskaya-daborovikov-sw-3>en
Password:
msk-donskaya-daborovikov-sw-3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-daborovikov-sw-3(config)#int g0/2
msk-donskaya-daborovikov-sw-3(config-if)#switchport mode trunk
msk-donskaya-daborovikov-sw-3(config-if)#^Z
msk-donskaya-daborovikov-sw-3#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-daborovikov-sw-3#wr me
Building configuration...
[OK]
msk-donskaya-daborovikov-sw-3#
```

Figure 2: g0/2 mode trunk

sw-1 и sw-4 через интерфейсы Fa0/23

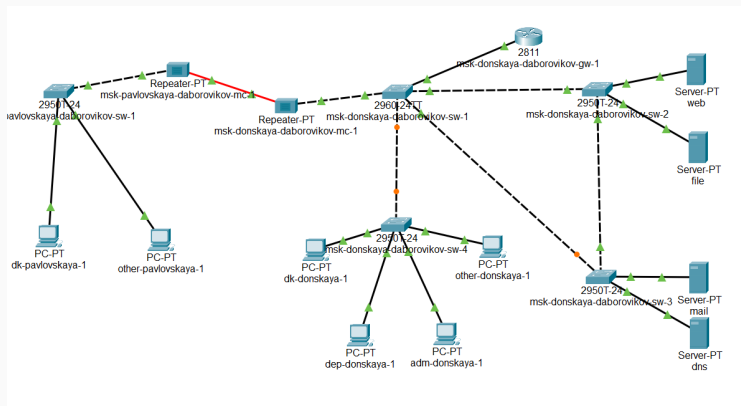


Figure 3: sw-1 и sw-4 через интерфейсы Fa0/23

sw-1 mode trunk

```
msk-donskaya-daborovikov-sw-1>en
Password:
msk-donskaya-daborovikov-sw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-daborovikov-sw-1(config)#int fa0/
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/23 (1), with msk-
donskaya-daborovikov-sw-4 FastEthernet0/23int fa0/
^
% Invalid input detected at '^' marker.

msk-donskaya-daborovikov-sw-1(config)#int fa0/23
msk-donskaya-daborovikov-sw-1(config-if)#switchport mode trunk

msk-donskaya-daborovikov-sw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to down

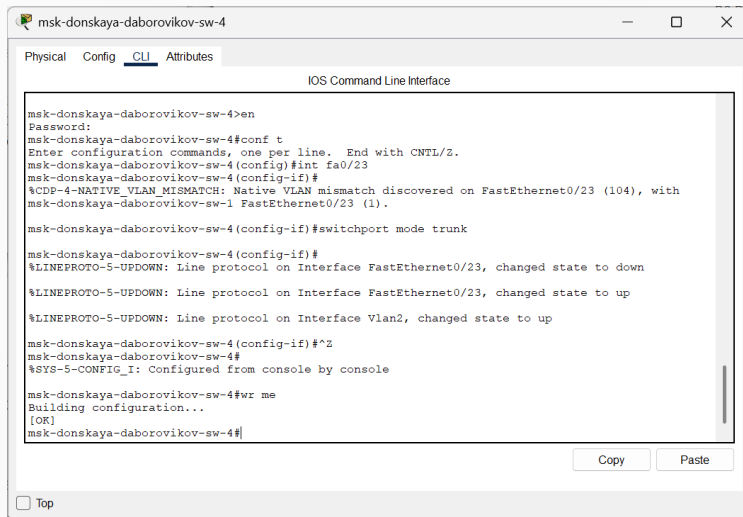
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to up

msk-donskaya-daborovikov-sw-1(config-if)#^Z
msk-donskaya-daborovikov-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-daborovikov-sw-1#wr me
Building configuration...
[OK]
msk-donskaya-daborovikov-sw-1#
```

Figure 4: sw-1 mode trunk

sw-4 mode trunk



The screenshot shows a window titled "msk-donskaya-daborovikov-sw-4" with tabs for "Physical", "Config", "CLI", and "Attributes". The "CLI" tab is active, displaying the "IOS Command Line Interface". The terminal text is as follows:

```
msk-donskaya-daborovikov-sw-4>en
Password:
msk-donskaya-daborovikov-sw-4#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-daborovikov-sw-4(config)#int fa0/23
msk-donskaya-daborovikov-sw-4(config-if)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/23 (104), with
msk-donskaya-daborovikov-sw-1 FastEthernet0/23 (1).

msk-donskaya-daborovikov-sw-4(config-if)#switchport mode trunk

msk-donskaya-daborovikov-sw-4(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to up

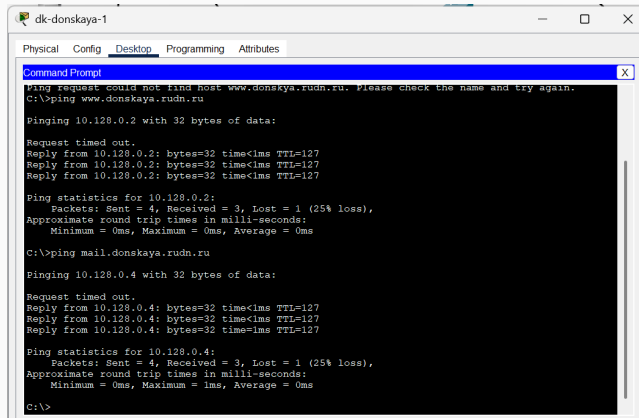
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan2, changed state to up

msk-donskaya-daborovikov-sw-4(config-if)#^Z
msk-donskaya-daborovikov-sw-4#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-daborovikov-sw-4#wr me
Building configuration...
[OK]
msk-donskaya-daborovikov-sw-4#
```

At the bottom right of the CLI window are "Copy" and "Paste" buttons. At the bottom left of the window is a checkbox labeled "Top".

Figure 5: sw-4 mode trunk



```
dk-donskaya-1
Physical Config Desktop Programming Attributes
Command Prompt
ping request could not find host www.donskaya.rudn.ru. Please check the name and try again.
C:\>ping www.donskaya.rudn.ru

Pinging 10.128.0.2 with 32 bytes of data:

Request timed out.
Reply from 10.128.0.2: bytes=32 time<1ms TTL=127
Reply from 10.128.0.2: bytes=32 time<1ms TTL=127
Reply from 10.128.0.2: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.0.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping mail.donskaya.rudn.ru

Pinging 10.128.0.4 with 32 bytes of data:

Request timed out.
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time=1ms TTL=127

Ping statistics for 10.128.0.4:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Figure 6: ping

движение пакетов через sw-2

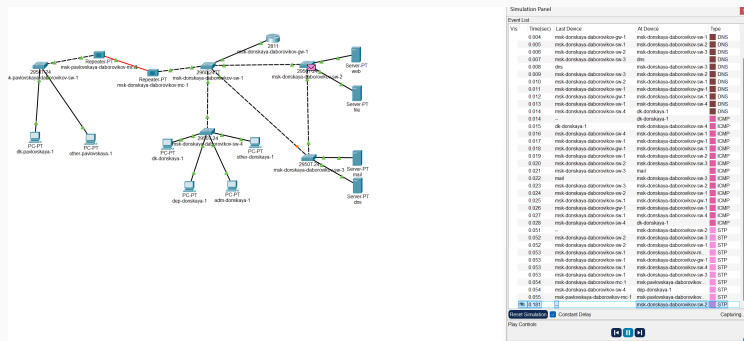
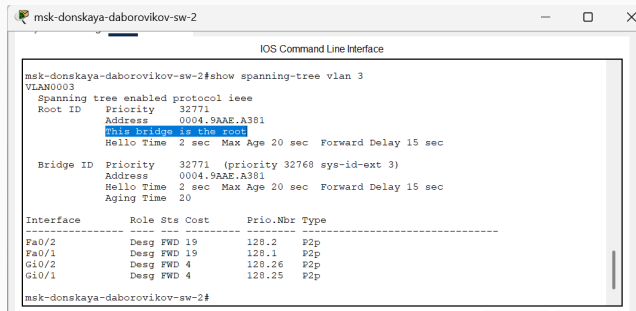


Figure 7: движение пакетов через sw-2

Состояние протокола STP



```
msk-donskaya-daborovikov-sw-2#show spanning-tree vlan 3
VLAN0003
  Spanning tree enabled protocol ieee
  Root ID    Priority    32771
             Address     0004.9AAE.A381
             This bridge is the root
             Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec

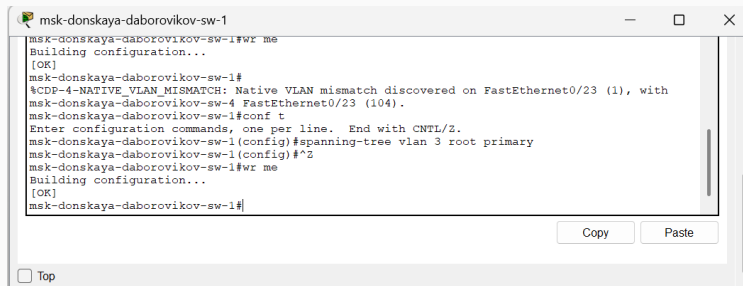
  Bridge ID  Priority    32771 (priority 32768 sys-id-ext 3)
             Address     0004.9AAE.A381
             Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time 20

Interface    Role Sts Cost      Prio.Nbr Type
-----
Fa0/2        Desg FWD 19        128.2   P2p
Fa0/1        Desg FWD 19        128.1   P2p
Gi0/2        Desg FWD 4         128.26  P2p
Gi0/1        Desg FWD 4         128.25  P2p

msk-donskaya-daborovikov-sw-2#
```

Figure 8: Состояние протокола STP

sw-1 в качестве корневого коммутатора STP



```
msh-donskaya-daborovikov-sw-1
Building configuration...
[OK]
msh-donskaya-daborovikov-sw-1#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/23 (1), with
msh-donskaya-daborovikov-sw-4 FastEthernet0/23 (104).
msh-donskaya-daborovikov-sw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msh-donskaya-daborovikov-sw-1(config)#spanning-tree vlan 3 root primary
msh-donskaya-daborovikov-sw-1(config)#^Z
msh-donskaya-daborovikov-sw-1#wr me
Building configuration...
[OK]
msh-donskaya-daborovikov-sw-1#
```

Copy Paste

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Figure 9: sw-1 в качестве корневого коммутатора STP

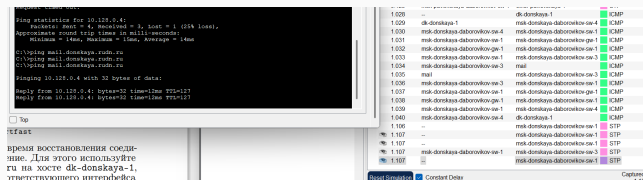


Figure 10: mail через sw-3

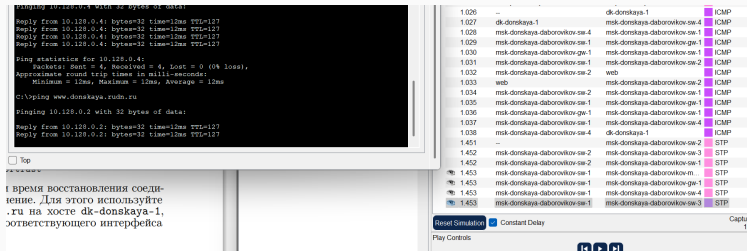
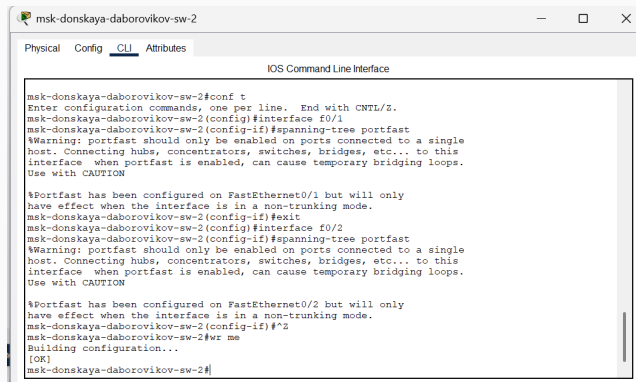


Figure 11: web через sw-2

Настройка режима Portfast sw-2



The screenshot shows a web-based interface for a network switch named 'msk-donskaya-daborovikov-sw-2'. The 'CLI' tab is selected, displaying the 'IOS Command Line Interface'. The terminal shows the following commands and output:

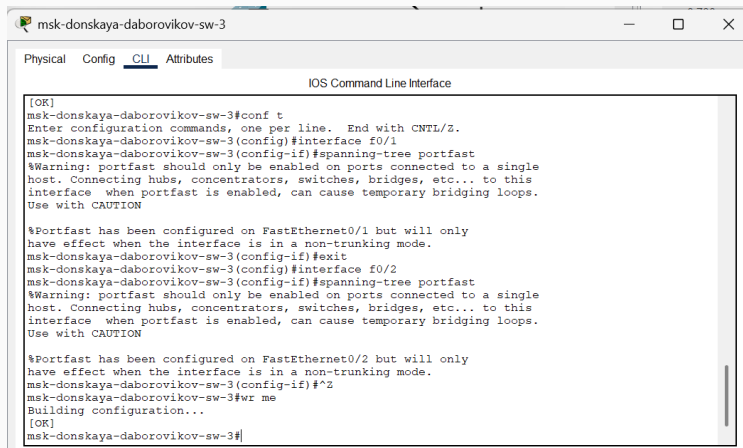
```
msk-donskaya-daborovikov-sw-2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-daborovikov-sw-2(config)#interface f0/1
msk-donskaya-daborovikov-sw-2(config-if)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/1 but will only
have effect when the interface is in a non-trunking mode.
msk-donskaya-daborovikov-sw-2(config-if)#exit
msk-donskaya-daborovikov-sw-2(config)#interface f0/2
msk-donskaya-daborovikov-sw-2(config-if)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/2 but will only
have effect when the interface is in a non-trunking mode.
msk-donskaya-daborovikov-sw-2(config-if)#^Z
msk-donskaya-daborovikov-sw-2#wr me
Building configuration...
[OK]
msk-donskaya-daborovikov-sw-2#
```

Figure 12: Настройка режима Portfast sw-2

Настройка режима Portfast sw-3



The screenshot shows a web-based CLI interface for a switch named 'msk-donskaya-daborovikov-sw-3'. The 'CLI' tab is selected. The interface displays the following commands and messages:

```
[OK]
msk-donskaya-daborovikov-sw-3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-daborovikov-sw-3(config)#interface f0/1
msk-donskaya-daborovikov-sw-3(config-if)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/1 but will only
have effect when the interface is in a non-trunking mode.
msk-donskaya-daborovikov-sw-3(config-if)#exit
msk-donskaya-daborovikov-sw-3(config)#interface f0/2
msk-donskaya-daborovikov-sw-3(config-if)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/2 but will only
have effect when the interface is in a non-trunking mode.
msk-donskaya-daborovikov-sw-3(config-if)#^Z
msk-donskaya-daborovikov-sw-3#wr me
Building configuration...
[OK]
msk-donskaya-daborovikov-sw-3#
```

Figure 13: Настройка режима Portfast sw-3

Отказоустойчивость протокола STP

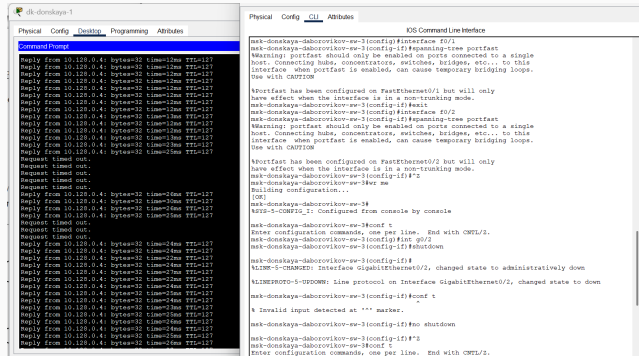
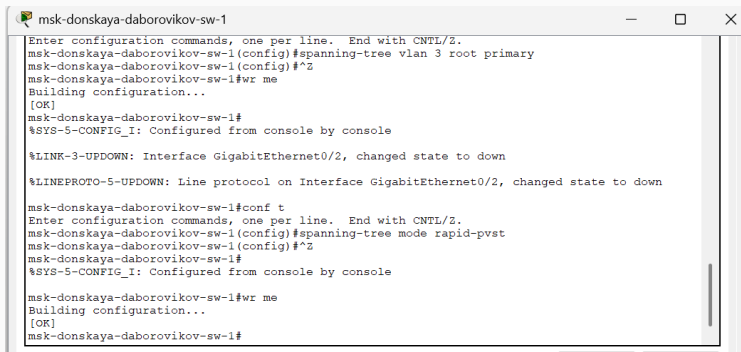


Figure 14: Отказоустойчивость протокола STP

Режим работы по протоколу Rapid PVST+



```
msk-donskaya-daborovikov-sw-1
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-daborovikov-sw-1(config)#spanning-tree vlan 3 root primary
msk-donskaya-daborovikov-sw-1(config)#^Z
msk-donskaya-daborovikov-sw-1#wr me
Building configuration...
[OK]
msk-donskaya-daborovikov-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

%LINK-3-UPDOWN: Interface GigabitEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down

msk-donskaya-daborovikov-sw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-daborovikov-sw-1(config)#spanning-tree mode rapid-pvst
msk-donskaya-daborovikov-sw-1(config)#^Z
msk-donskaya-daborovikov-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-daborovikov-sw-1#wr me
Building configuration...
[OK]
msk-donskaya-daborovikov-sw-1#
```

Figure 15: Режим работы по протоколу Rapid PVST+

Отказоустойчивость протокола Rapid PVST+

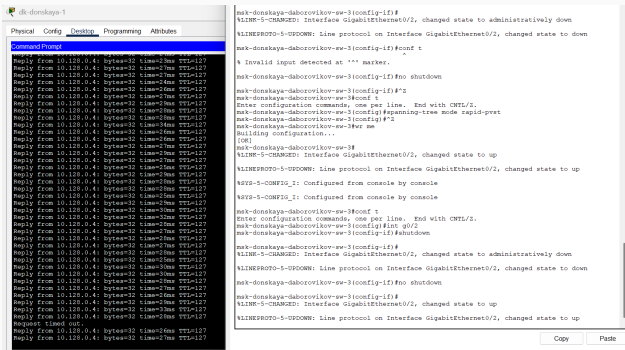


Figure 16: Отказоустойчивость протокола Rapid PVST+

Логическая схема локальной сети с агрегированным соединением

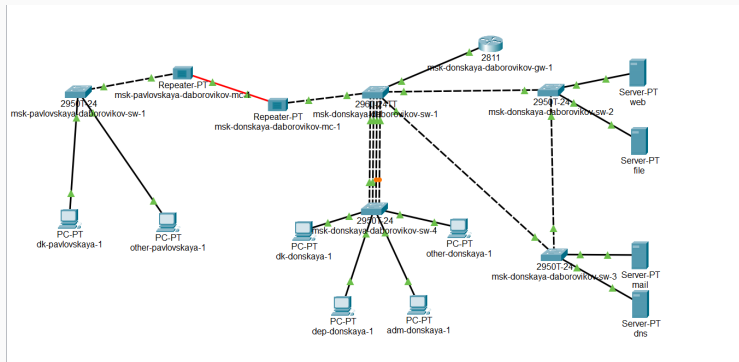
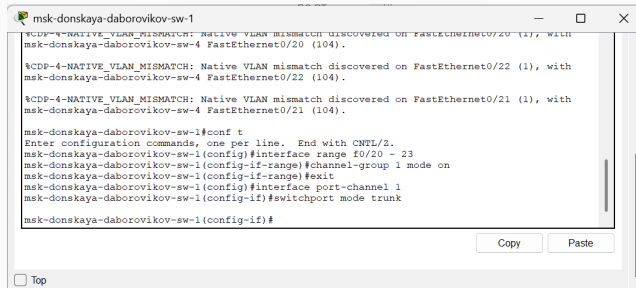


Figure 17: Логическая схема локальной сети с агрегированным соединением

sw-1 агрегирование каналов (режим EtherChannel)



```
mzk-donskaya-daborovikov-sw-1
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/20 (1), with
mzk-donskaya-daborovikov-sw-4 FastEthernet0/20 (104).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/22 (1), with
mzk-donskaya-daborovikov-sw-4 FastEthernet0/22 (104).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/21 (1), with
mzk-donskaya-daborovikov-sw-4 FastEthernet0/21 (104).

mzk-donskaya-daborovikov-sw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
mzk-donskaya-daborovikov-sw-1(config)#interface range f0/20 - 23
mzk-donskaya-daborovikov-sw-1(config-if-range)#channel-group 1 mode on
mzk-donskaya-daborovikov-sw-1(config-if-range)#exit
mzk-donskaya-daborovikov-sw-1(config)#interface port-channel 1
mzk-donskaya-daborovikov-sw-1(config-if)#switchport mode trunk

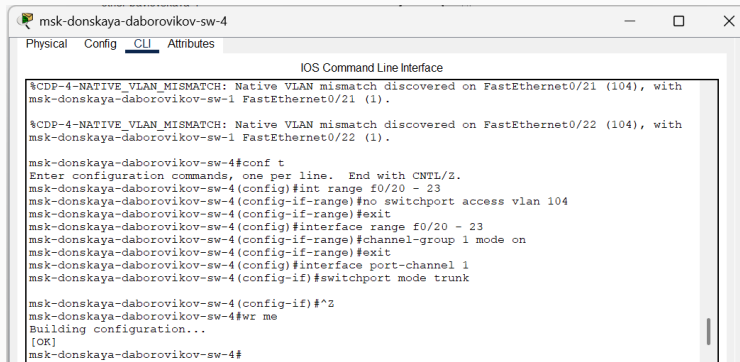
mzk-donskaya-daborovikov-sw-1(config-if)#
```

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Figure 18: sw-1 агрегирование каналов (режим EtherChannel)

sw-4 агрегирование каналов (режим EtherChannel)



The screenshot shows a terminal window titled "msk-donskaya-daborovikov-sw-4" with tabs for Physical, Config, CLI, and Attributes. The CLI tab is active, displaying the "IOS Command Line Interface". The output shows two CDP error messages about VLAN mismatches on interfaces FastEthernet0/21 and FastEthernet0/22. Then, the user enters configuration mode and configures an EtherChannel group 1 with member interfaces f0/20 through f0/23. The configuration is saved and the prompt returns to user mode.

```
msk-donskaya-daborovikov-sw-4#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/21 (104), with
msk-donskaya-daborovikov-sw-1 FastEthernet0/21 (1).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/22 (104), with
msk-donskaya-daborovikov-sw-1 FastEthernet0/22 (1).

msk-donskaya-daborovikov-sw-4#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-daborovikov-sw-4(config)#int range f0/20 - 23
msk-donskaya-daborovikov-sw-4(config-if-range)#no switchport access vlan 104
msk-donskaya-daborovikov-sw-4(config-if-range)#exit
msk-donskaya-daborovikov-sw-4(config)#interface range f0/20 - 23
msk-donskaya-daborovikov-sw-4(config-if-range)#channel-group 1 mode on
msk-donskaya-daborovikov-sw-4(config-if-range)#exit
msk-donskaya-daborovikov-sw-4(config)#interface port-channel 1
msk-donskaya-daborovikov-sw-4(config-if)#switchport mode trunk

msk-donskaya-daborovikov-sw-4(config-if)#^Z
msk-donskaya-daborovikov-sw-4#wr me
Building configuration...
[OK]
msk-donskaya-daborovikov-sw-4#
```

Figure 19: sw-4 агрегирование каналов (режим EtherChannel)

В ходе выполнения лабораторной работы я изучил возможности протокола STP и его модификаций по обеспечению отказоустойчивости сети, агрегированию интерфейсов и перераспределению нагрузки между ними.